

Predicting Anime TV Show Scores from MyAnimeList

CS 539: Machine Learning

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Overview

Project Premise

This project aims to predict the popularity of anime shows by analyzing a comprehensive dataset from Kaggle. This dataset, derived from MyAnimeList, comprises 12,294 anime shows, each characterized by 35 distinct features such as title, genre, studio, source, rating, and episode count. The key metric, a popularity score ranging from 0 to 100,000, indicates the number of MyAnimeList users who have added the show to their list.

Data Preparation and Analysis

The initial step involved cleansing and preprocessing the dataset by eliminating irrelevant columns and rows with missing data. Remaining gaps were filled using column mean values. For analytical convenience, categorical variables were quantified based on the average popularity scores of shows within each category. Similarly, numerical variables underwent a logarithmic transformation. The dataset was then partitioned into training (80%) and testing (20%) subsets for model training and evaluation, respectively. Model performance was assessed using 5-fold cross-validation, focusing on explained variance (R^2) as the primary metric, where higher R^2 signifies superior model accuracy.

Model Implementation and Results

Several models were employed, including Linear Regression, K-Nearest Neighbors (KNN), Neural Network, Decision Tree, Random Forest, and Polynomial Regression. These models were tested against three feature sets: numerical only, categorical only, and a combined set. The table below encapsulates the findings:

ML Model	X_{num}	X_{cat}	X
Linear Regression	0.67	0.42	0.73
KNN	0.60	0.34	0.66
Neural Network	0.66	0.26	0.70
Decision Tree	0.26	-0.09	0.43
Random Forest	0.57	0.38	0.70
Polynomial Regression	0.67	0.43	0.72

The combined feature set consistently yielded the most effective results across most models, underscoring the importance of integrating both numerical and categorical data for a holistic analysis. The varying model performance across different feature sets highlighted the significance of model choice relative to the nature of the data. Notably, models traditionally proficient with categorical data, like Decision Trees and KNN, did not meet expectations, hinting at potential feature representation or parameterization issues.

Further Considerations and Conclusion

Subsequent attempts at optimizing the models through feature engineering and hyperparameter adjustments have been largely inconsequential. The dataset's age (at least 5 years old) suggests that incorporating more recent data could enhance the analysis. Moreover, the methodology for translating categorical variables into meaningful scores may be overly simplistic. Although efforts were made to develop a more sophisticated model, the improvements, if any, were minimal.

Summary

In this project, machine learning techniques were employed to predict anime show popularity using a Kaggle dataset from MyAnimeList, encompassing 12,294 shows and 35 features, including genre, rating, and studio. The data underwent preprocessing, such as imputing missing values and log-transforming numerical variables, followed by conversion of categorical data into numerical scores. Then, an 80-20 split was utilized for training and testing sets when training models like Linear Regression, Neural Networks, Decision Trees, Random Forests, and Polynomial Regression. Evaluation was based on 5-fold cross-validation focusing on MSE. The project highlighted the significance of combining numerical and categorical features for improved model performance, though it also revealed the challenges in feature representation and the limitations of the dataset's age.

Project Description

Motivation

The motivation behind this project stems from a deep interest in the anime industry and a desire to understand the factors that contribute to an anime's popularity. As an anime enthusiast, the project was driven by a personal quest to discover what makes certain anime shows resonate with audiences. This motivation extended to a professional curiosity about the market dynamics within the anime industry, particularly how certain elements like genre, studio, and production values can influence a show's success.

Objectives

The primary objective of this project was to develop a machine learning model capable of predicting the popularity scores of anime shows on MyAnimeList. The goals included:

1. Identifying key factors that significantly impact an anime's popularity.
2. Analyzing the relationship between these factors and the anime's popularity score.
3. Developing a predictive model that could accurately forecast the popularity scores based on identified features.

Dataset

The dataset used in this project was sourced from [Azathoth's MyAnimeList dataset](#). It included information on 12,294 anime shows, encompassing a wide range of features such as genre, studio, source material, and user ratings. Each show was characterized by 35 distinct attributes, providing a comprehensive dataset for analysis. The size and diversity of this dataset offered a robust foundation for the machine learning models to uncover meaningful patterns and relationships.

Methods

The methodology for this project involved a systematic approach to data preparation, feature engineering, model training, and evaluation.

Data Preparation

The dataset, sourced from Kaggle's "MyAnimeList" collection, was initially cleaned by removing irrelevant columns and handling rows with missing data. This process included imputing missing values with column mean values where appropriate. The dataset was then partitioned into training (80%) and testing (20%) subsets. This split ensured a robust evaluation of the models on unseen data.

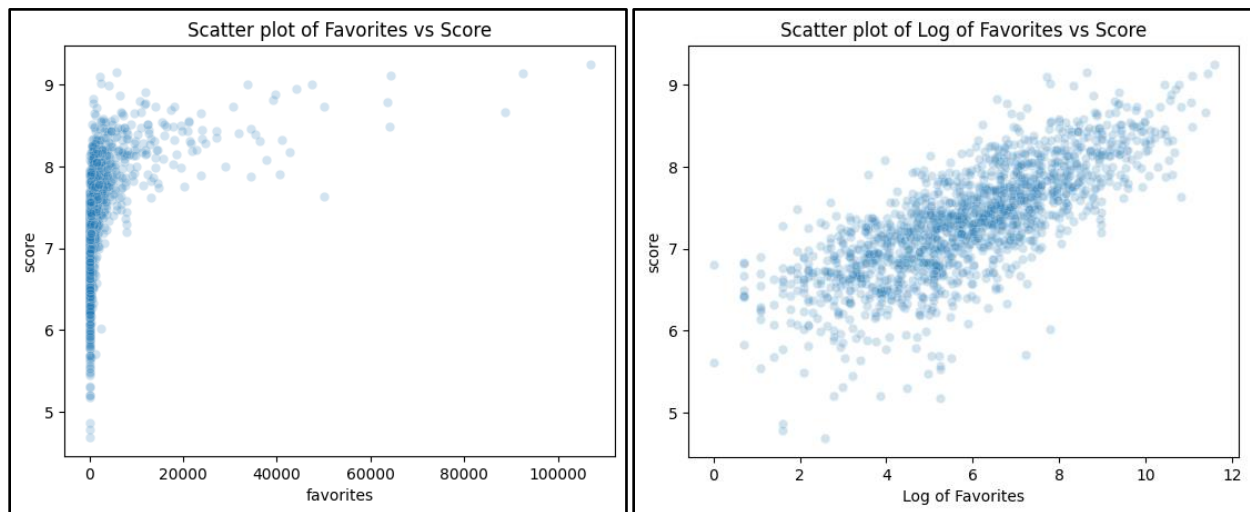
Feature Engineering

Numerical Features

This feature set (denoted X_{num}), contains the features $\log_{\text{members}}$, $\log_{\text{popularity}}$, and $\log_{\text{favorites}}$. A logarithmic transformation was applied to these variables to linearize their distribution and improve model performance.

For each numerical value N , $\log(N + 1)$ was applied to it. The $+1$ is there to prevent any instance of $\log(0)$, which is undefined.

For example, a favorites count of 574 would be used to create a new feature called $\log_{\text{favorites}}$ feature with a value of $\log(574 + 1)$.



Categorical Features

This feature set (denoted X_{cat}), contains the features $\text{mean}_{\text{genre}}$, $\text{mean}_{\text{studio}}$, $\text{mean}_{\text{producer}}$, and $\text{mean}_{\text{licensor}}$. Attributes such as genre, studio, producer, and licensor (GSPL) were encoded into numerical formats. This involved computing the average popularity scores of shows within each category, thereby converting these categorical variables into quantitative measures.

For example, a show with the genres Fantasy and Sci-Fi would have $\text{mean}_{\text{genre}}$ feature with the value $\text{mean}(\text{mean}(F) + \text{mean}(S))$, where...

- F = mean score of all Fantasy anime tv shows
- S = mean score of all Sci-Fi anime tv shows

Most importantly, the approach to convert these categorical variables into numerical features is to better reflect the relationship these have with the actual score. I have considered some other commonly used approaches for handling categorical variables that I do not think really apply in this situation (in a semantic/philosophical sense) and/or would be practically inefficient.

One-hot encoding

Given that there are about 500 producers and 180 studios, this would result in a very sparse table where only a relatively small number of columns would be relevant for a given show.

The set of GSPL scraped in the dataset probably also does not overlap, if we do want the model to consider other shows outside the dataset, where they could be a part of a new genre and are made by new studios.

Ordinal Encoding

This may imply a hierarchy amongst GSPL (i.e., X studio is better than Y studio), which would semantically misrepresent what people care about these categories.

For genres, that is up to personal preference on the individual level, and thus not generalizable enough to easily encode in an ML model to predict the aggregate scoring of shows based on a relatively wide audience.

For studios/producers/licensors, people really only care about the shows they've worked on, whose information could be somewhat captured with ordinal encoding, but this method seems overly simplistic.

Embedding

This is a common technique used in NLP problems that concerns the meaning of those words. For this problem, I do not care exactly about the names of each GSPL, but (as mentioned before) the shows associated with each GSPL.

With that said, the current implementation of converting GSPL names into score metrics that are representative of their related shows is overly simplistic for what I eventually want to attain, but this approach can be adapted to accept more sophisticated implementations.

- e.g.) if a studio is known to make better shows of genre A than genre B, the scores the former should be weighed more than the scores of the latter.
- e.g.) a studio should have their more recent shows weighed more than their earlier shows.

Combined Feature Set

The model was also tested on a combination of both numerical and categorical features to examine the effect of integrating diverse data types.

This feature engineering process was critical for preparing the data for machine learning algorithms, as it transformed the dataset into a format that could be efficiently processed and analyzed.

Model Training and Evaluation

In this project, our model training and evaluation were executed with a focus on comprehensiveness and rigor, employing a variety of machine learning algorithms and performance metrics. The chosen algorithms, each with its strengths and ideal use cases, were Linear Regression, K-Nearest Neighbors (KNN), Neural Networks, Decision Trees, Random Forests, and Polynomial Regression.

Models

Linear Regression was selected for its simplicity and effectiveness in predicting continuous outcomes. It serves as a baseline model, providing a point of reference for the performance of more complex algorithms. Linear Regression's interpretability was also a factor, allowing for an easy understanding of how each feature influences the predicted score.

K-Nearest Neighbors (KNN), a non-parametric algorithm, was used to explore how a more instance-based approach would fare in this context. KNN's performance in this project was particularly indicative of the dataset's structure and feature space since it relies on the proximity of data points to make predictions.

Neural Networks offered a more sophisticated and layered approach to prediction, capable of capturing complex, non-linear relationships in the data. The architecture of the neural network was fine-tuned to balance between underfitting and overfitting, considering aspects like the number of layers and neurons in each layer.

Decision Trees provided a more visual and intuitive way of understanding the data's decision-making process. However, their tendency to overfit made them less reliable, necessitating the use of **Random Forests**. As an ensemble of Decision Trees, Random Forests mitigate overfitting through bagging and feature randomness, offering a more robust and generalized model.

Polynomial Regression extended the capabilities of Linear Regression to capture non-linear relationships. By fitting polynomial features, this model was tested to see if the relationship between features and the target variable exhibited a higher degree of complexity than what a simple linear model could capture.

Performance Metrics

Each model was subjected to rigorous evaluation using a suite of performance metrics:

- **Mean Square Error (MSE)** and **Root Mean Squared Error (RMSE)** provided insights into the average magnitude of the models' prediction errors, with RMSE being particularly sensitive to large errors due to its squaring of the residuals.
- **Mean Absolute Error (MAE)** offered a straightforward interpretation of average error magnitude, unaffected by the direction of the error, making it a reliable measure of model performance.
- **R² (Explained Variance)** was crucial for understanding the proportion of variance in the anime popularity scores that our models could explain. A higher R² indicated a model that could account for a greater fraction of the variance.

The evaluation was not limited to these static metrics. Employing **5-fold cross-validation** provided a more robust assessment, ensuring that the model's performance was consistent across different subsets of the data. This approach mitigated the risk of overfitting, as each model was trained and validated on five different splits of the dataset.

The diversity of models tested, along with a comprehensive set of evaluation metrics, ensured a thorough investigation into the dataset's underlying patterns. This approach provided a nuanced understanding of the factors driving anime popularity, allowing us to identify the most effective model for this specific application.

Conclusion

The culmination of this project's journey in machine learning to predict anime popularity on MyAnimeList has led to several significant insights and conclusions. By applying various machine learning models to a dataset encompassing 12,294 anime shows, the project not only achieved its primary objective of developing a predictive model but also uncovered deeper understandings of the factors influencing anime popularity.

Key Insights and Model Performance

Integration of Numerical and Categorical Data

The combined use of both numerical and categorical features in our models consistently yielded more accurate predictions. This suggests that a comprehensive approach, considering various aspects like genre, studio, and user engagement metrics, is crucial for a holistic understanding of anime popularity.

Model Variability

The performance of different models varied significantly. Linear and Polynomial Regression models showed robustness across all feature sets, indicating their effectiveness in capturing both linear and more complex relationships. On the other hand, models like KNN and Neural Networks, while performing moderately well with numerical data, struggled with categorical data. This disparity points to the complexity and nuanced nature of the data, emphasizing the need for careful consideration in feature engineering and model selection.

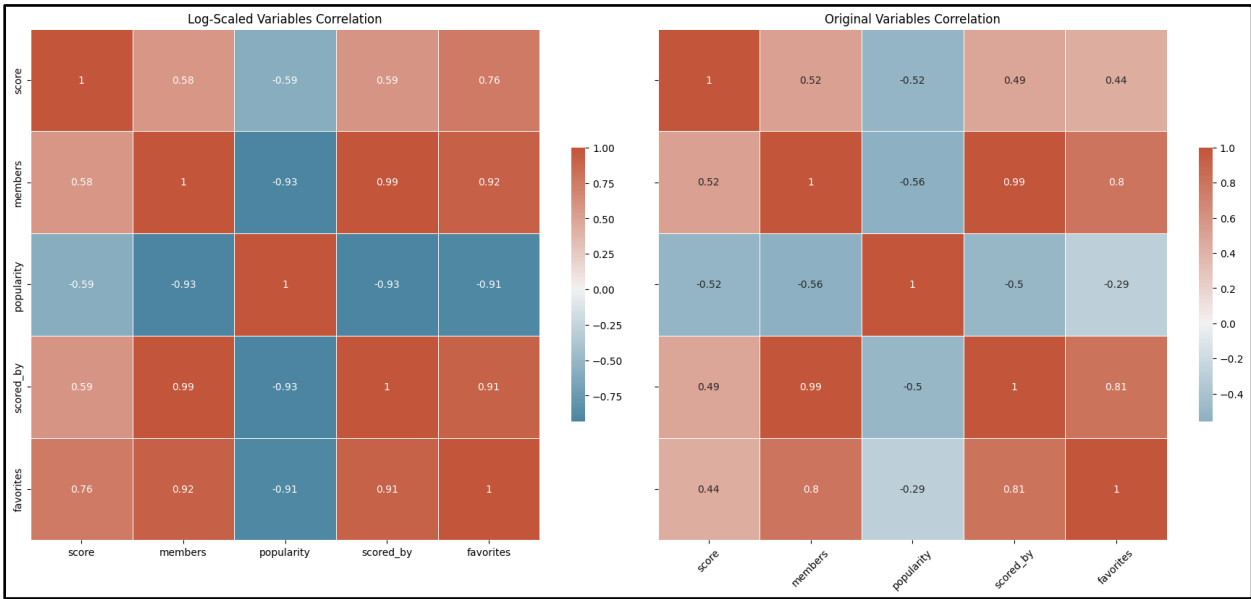
Challenges in Feature Representation

The project highlighted the intricacies involved in representing categorical data effectively. Models like Decision Trees and Random Forests, typically strong with categorical data, did not perform as well as expected. This suggests potential areas for improvement in how these categorical variables are encoded and utilized in the models.

Visualizations and Data Trends

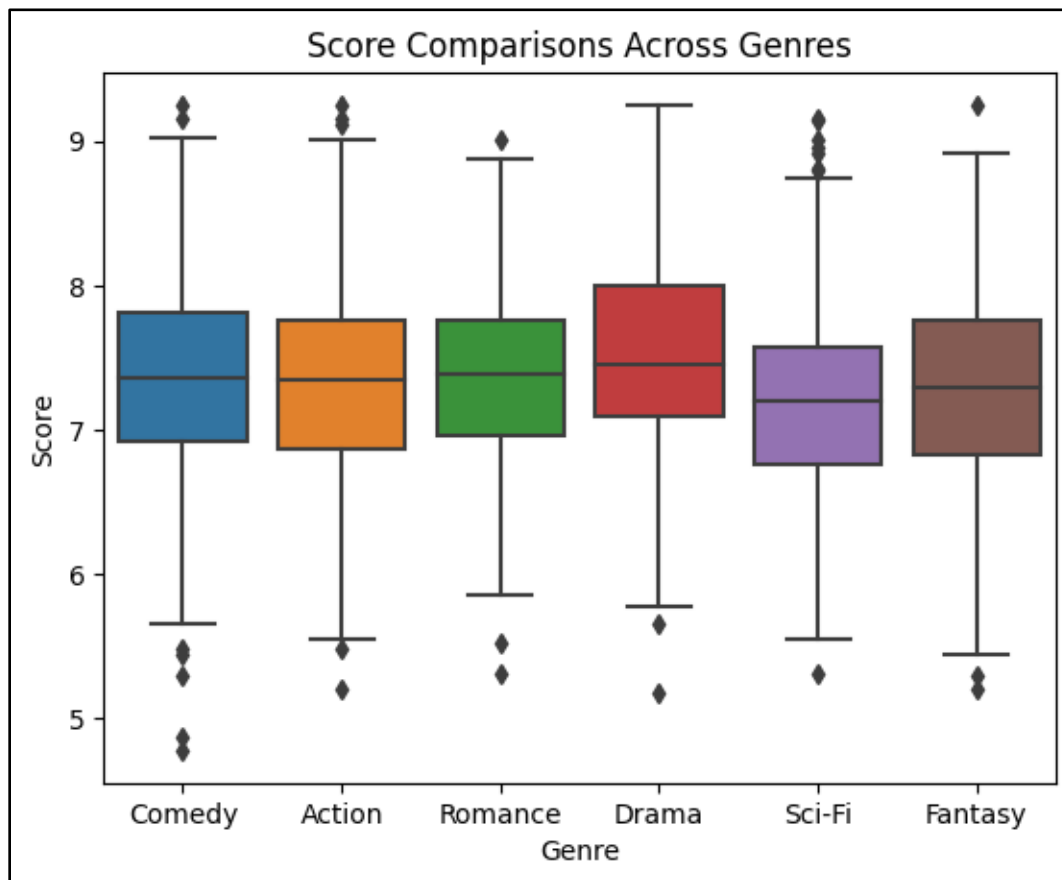
Several visualizations were instrumental in guiding the project's analysis and conclusions.

Correlation Heatmaps



These provided clear insights into how different variables interacted with the target variable (popularity score). The logarithmic transformations applied to numerical features like favorites count and member count helped in enhancing these correlations, as evident from the increase in Pearson coefficients.

Genre-based Score Distributions



Box-and-whisker plots for different genres revealed variations in scoring patterns across genres, with genres like Drama scoring higher on average. This highlighted the potential of genre as a predictive feature and informed the decision to compute mean scores for genres in feature engineering.

Limitations and Future Directions

Despite the project's achievements, certain limitations were identified.

Dataset Age

The dataset's age (at least 5 years old) limits the applicability of the model to current trends. Incorporating more recent data could significantly enhance the model's relevance and accuracy.

Simplistic Categorical Feature Conversion

The approach to convert categorical variables like genre and studio into numerical scores was somewhat rudimentary. Future work could focus on developing more sophisticated methods for this conversion, potentially improving the model's performance.

Scope for Optimization

Although attempts at further optimization through advanced feature engineering and hyperparameter tuning were made, the improvements were minimal. This suggests a potential ceiling in the model's performance given the current dataset and feature engineering approach.